DLMS Quick Start Guide

5-Minute Setup

Prerequisites Check

Before starting, ensure you have:

- ✅ Python 3.7+ installed (`python --version`)
- ✅ MySQL Server installed and running
- ✅ pip package manager available

Step-by-Step Setup

1. Install MySQL (if not installed)

Windows: Download from https://dev.mysql.com/downloads/mysql/ **Mac:** `brew install mysql` **Linux:** `sudo apt-get install mysql-server`

Start MySQL service:

- **Windows:** Start MySQL service from Services
- **Mac/Linux:** `mysql.server start` or `sudo systemctl start mysql`

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2. Create Project Structure

```bash
mkdir dlms_project
cd dlms_project

# Create folders
mkdir templates
mkdir static
mkdir static/css
mkdir static/uploads
mkdir static/uploads/videos
```

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3. Install Python Dependencies

```bash
pip install flask flask-mysqldb werkzeug
```

Or use requirements.txt:

```bash
pip install -r requirements.txt
```

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4. Setup Database

Open MySQL command line:

```bash
mysql -u root -p
```

Then execute:

```sql
CREATE DATABASE dlms_db;
USE dlms_db;

-- Copy and paste all SQL from database.sql file
-- Or run: source /path/to/database.sql
```

Note: The SQL file includes default user accounts with hashed passwords.

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5. Configure Application

Open `app.py` and update these lines:

```python
```

```
# Line 8 - Change secret key
app.secret_key = 'your_random_secret_key_here'

# Lines 11-14 - Update MySQL credentials
app.config['MYSQL_HOST'] = 'localhost'
app.config['MYSQL_USER'] = 'root'
app.config['MYSQL_PASSWORD'] = 'your_mysql_password'
app.config['MYSQL_DB'] = 'dlms_db'
```

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6. File Placement

Ensure files are in correct locations:

```
dlms_project/
│
├── app.py                    # Main application file
├── database.sql              # Database schema
├── requirements.txt          # Python dependencies
│
├── templates/                # All HTML files go here
│   ├── base.html
│   ├── index.html
│   ├── login.html
│   └── ... (all other .html files)
│
└── static/
    ├── css/
    │   └── style.css         # Main stylesheet
    └── uploads/
        └── videos/           # Video storage (created automatically)
```

---

7. Start the Application

```bash
cd dlms_project
python app.py
```

You should see:

```
* Running on http://127.0.0.1:5000
* Debug mode: on
```

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8. Access the System

Open your browser and go to: **http://127.0.0.1:5000**

Default Login Credentials

Administrator

- **Email:** admin@dlms.com
- **Password:** admin123

Lecturer

- **Email:** lecturer@dlms.com
- **Password:** lecturer123

Student

- **Email:** student@dlms.com
- **Password:** student123

First Time Usage Flow

For Administrators (Test First)

1. Login with admin credentials
2. Navigate to "Manage Users"
3. Create a new student account
4. Create a new lecturer account
5. Test activation/deactivation

For Lecturers

1. Login with lecturer credentials
2. Create a course:

- Go to "Create Course"
- Enter: "Web Development 101"
- Description: "Introduction to HTML, CSS, and JavaScript"
- Submit

3. Upload a lecture:
    - Go to "Upload Lecture"
    - Title: "HTML Basics"
    - Description: "Learn HTML fundamentals"
    - Select your course
    - Upload a small test video (MP4)
    - Submit

For Students

1. Login with student credentials
2. Enroll in course:
    - Go to "Enroll in Course"
    - Click "Enroll Now" on "Web Development 101"
3. Access lecture:
    - Go to "My Courses"
    - Click "View Lectures"
    - Click "Watch Lecture"
    - Video should play

Common Issues & Solutions

Issue 1: "Module not found" error

Solution: Install required packages

```bash
pip install flask flask-mysqldb werkzeug
```

Issue 2: "Access denied for user" (MySQL)

Solution: Check MySQL credentials in app.py match your MySQL setup

Issue 3: "Can't connect to MySQL server"

Solution: Ensure MySQL service is running

```bash
# Check if MySQL is running
mysql -u root -p

# If not, start it
mysql.server start  # Mac/Linux
# Or start MySQL service in Windows Services
```

Issue 4: Video upload fails

Solution: Check:

- Video file is under 100MB
- Video format is MP4, AVI, MOV, WMV, or MKV
- uploads/videos folder exists and is writable

Issue 5: Import error for MySQLdb

Solution: Install MySQL client

```bash
# Mac
brew install mysql

# Ubuntu/Debian
sudo apt-get install python3-dev default-libmysqlclient-dev build-essential

# Windows
pip install mysqlclient
```

Issue 6: Page not found (404)

Solution: Ensure:

- Flask app is running
- URL is exactly http://127.0.0.1:5000
- All template files are in templates/ folder

Issue 7: CSS not loading

Solution:

- Ensure style.css is in static/css/
- Hard refresh browser (Ctrl+F5 or Cmd+Shift+R)
- Check Flask is serving static files

Testing Your Installation

Quick Test Checklist

1. ✅ Can access homepage
2. ✅ Can login as administrator
3. ✅ Can create a new user
4. ✅ Can login as lecturer
5. ✅ Can create a course
6. ✅ Can upload a video
7. ✅ Can login as student
8. ✅ Can enroll in course
9. ✅ Can watch video

If all these work, your installation is successful!

For Demo/Presentation Preparation

1 Day Before

- [ ] Test all features thoroughly
- [ ] Prepare sample courses and videos
- [ ] Create additional test users
- [ ] Take screenshots of key features
- [ ] Print documentation

2 Hours Before

- [ ] Start MySQL service
- [ ] Start Flask application
- [ ] Open multiple browser tabs:

- Tab 1: Admin logged in
- Tab 2: Lecturer logged in
- Tab 3: Student logged in

☐ Test video playback
☐ Prepare backup video files

During Presentation

- Keep Flask terminal visible (shows professionalism)
- Have database client ready to show schema
- Demonstrate in order: Admin → Lecturer → Student
- Show security features last (access control demo)

File Download Checklist

Ensure you have all these files:

Core Application Files

☐ app.py
☐ database.sql
☐ requirements.txt
☐ README.md

Template Files (15 files)

☐ base.html
☐ index.html
☐ login.html
☐ admin_dashboard.html
☐ admin_users.html
☐ admin_create_user.html
☐ admin_courses.html
☐ lecturer_dashboard.html
☐ lecturer_courses.html
☐ lecturer_create_course.html
☐ lecturer_upload_lecture.html
☐ lecturer_course_lectures.html
☐ student_dashboard.html
☐ student_courses.html
☐ student_enroll.html

- [ ] student_course_lectures.html
- [ ] student_view_lecture.html

Static Files

- [ ] static/css/style.css

Documentation Files

- [ ] Testing_Documentation.md
- [ ] Presentation_Guide.md
- [ ] Quick_Start_Guide.md

Getting Sample Video Files

For testing, you need video files under 100MB in MP4, AVI, MOV, WMV, or MKV format.

Option 1: Use existing videos

- Short clips from your phone
- Downloaded educational videos
- Screen recordings

Option 2: Download free sample videos

- https://sample-videos.com/
- https://file-examples.com/index.php/sample-video-files/

Option 3: Create simple test video

- Use OBS Studio (free) to record screen
- Keep under 5 minutes for quick testing
- Save as MP4 format

Final Checklist Before Defense

Technical Readiness

- [ ] Application runs without errors
- [ ] Database populated with demo data
- [ ] At least 2 courses created with videos
- [ ] Multiple test users in each role

- [ ] All videos play correctly
- [ ] No Python warnings in console

Documentation Readiness

- [ ] README printed or accessible
- [ ] Test documentation reviewed
- [ ] Presentation notes ready
- [ ] Database schema diagram prepared
- [ ] System architecture diagram ready

Presentation Readiness

- [ ] Laptop fully charged
- [ ] Power adapter packed
- [ ] Backup on USB drive
- [ ] Browser bookmarks set
- [ ] Demo script reviewed
- [ ] Team roles assigned
- [ ] Questions anticipated

Need Help?

Common Commands Reference

Start MySQL:

```bash
mysql.server start          # Mac/Linux
# Or use Services on Windows
```

Start Flask App:

```bash
python app.py
```

Check Python Version:

```bash
python --version
```

Check Installed Packages:

```bash
pip list
```

Access MySQL CLI:

```bash
mysql -u root -p
```

Check if Port 5000 is Free:

```bash
# Mac/Linux
lsof -i :5000

# Windows
netstat -ano | findstr :5000
```

Success Indicators

Your system is ready when:

1. ✅ Flask starts without errors
2. ✅ All three user types can login
3. ✅ Lecturer can upload videos
4. ✅ Student can watch uploaded videos
5. ✅ Access control works (student can't access admin pages)
6. ✅ Videos stream smoothly

You're Ready!

Once all checklist items are complete, you have a fully functional Distance Learning Management System ready for demonstration and evaluation.

Good luck with your project defense! 🎓